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Presidential Attention Focusing in the Global Arena: The Impact of International Travel on Foreign Publics

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Presidents increasingly engage in public diplomacy, especially traveling to other nations. Often presidents target the public in the visited nation, with the aim of improving the opinion climate there and then using that improved opinion climate to further U.S. policy goals in the visited nation. To a degree, then, foreign travel resembles presidential domestic going public. This article tests one important linkage in the process from a presidential trip to foreign public opinion, whether trips can focus public opinion in the visited nation on the president. Analysis uses a pooled cross-sectional time series analysis of weekly Google Trends data for forty-two nations during Barack Obama's first term in office. Results suggest that presidential trips lead to impressive increases in searches on the president, an indication of increased attention to the president. The conclusion places the findings into two literatures, presidential going public and public diplomacy, and suggests directions for future research.

Presidents allocate a large proportion of their time and energy to foreign affairs. In pursuing their foreign policy goals, presidents (and their secretaries of state) have increased their travel to other nations since the end of the Second World War. Figure 1 plots the annual number of foreign trips by U.S. presidents from 1946 to 2011. From a handful of country visits during the mid-twentieth century, by the 2000s presidents routinely visit one and a half dozen nations per year, a sixfold increase from the 1950s. Secretaries of state are globally even more peripatetic than presidents, with an average of fifty-one nations visited per year in the 2000s, compared to a dozen in the 1950s.¹

On these visits, presidents meet with foreign leaders, performing classic diplomatic activities such as attending formal negotiation meetings that are held in secret. But now

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1. Each visit to a foreign nation is counted separately, even if the president visited the nation two or more times. For the secretary of state totals, I do not count joint presidential–secretary trips. The secretary almost always accompanies the president on a foreign trip. The source for these data is the Office of the Historian of the State Department, <http://history.state.gov/>.

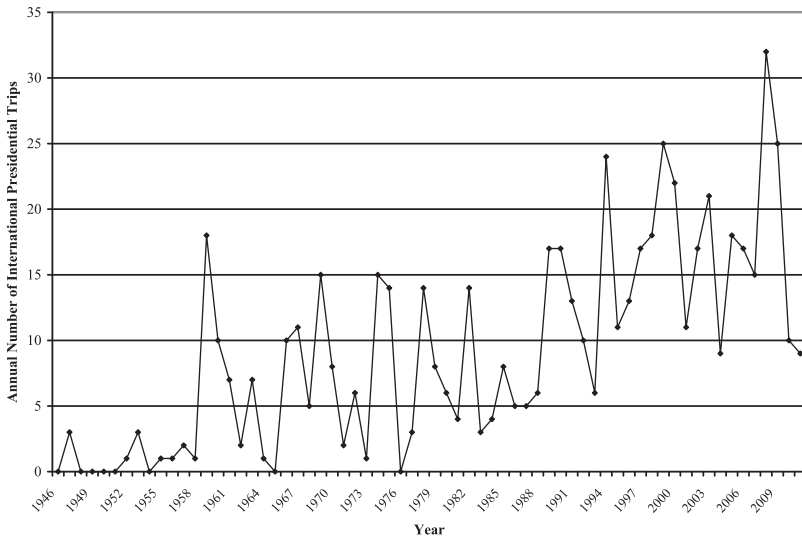


FIGURE 1. Annual Number of Countries Visited by the President, 1946-2011.

Source: Office of the Historian of the U.S. Department of State, <http://history.state.gov/departmentshistory>.

presidents also routinely appear in public when visiting other nations. For instance, presidents now commonly hold joint press conferences and public announcements with the leader of the foreign nation, give interviews with foreign journalists, visit locations of local symbolic importance, and directly address the local citizenry. As one prominent example, on August 3, 2009, President Obama held a U.S. campaign-style town hall meeting in Strasbourg, France, with French citizens. Goldsmith and Horiuchi (2009) have dubbed these public activities “high-level public diplomacy.”

High-level public diplomacy activities resemble “going public” techniques that presidents use in domestic politics and policy making (Kernell 1993). Domestic going public is predicated upon a two-step process. First, through public activities, like making speeches, presidents try to rally or mobilize public opinion around an issue. Then presidents use that activated public opinion to pressure members of Congress to enact their policy initiatives (Canes-Wrone 2006). A large literature has investigated the effects of presidential going public on domestic public opinion and Congress, finding mixed effects on both. In the domestic arena, presidential going public is far from consistently effective.²

In several respects, high-level public diplomacy efforts parallel those for domestic going public. First, like domestic going public, public diplomacy activities target foreign public opinion, with the aim of improving the image of the United States, opinion

2. The literature on going public and public opinion is extensive. See Edwards (2006, 2009) and Cohen (2010, 14-17) for useful reviews. The major studies of the impact of going public on Congress are Canes-Wrone (2005), Barrett (2004), and Powell and Schloyer (2003), but they differ, with Canes-Wrone suggesting conditional effectiveness, Barrett finding more general impacts, and Powell and Schloyer not finding going public to be effective in affecting voting in Congress.

concerning an international issue, and/or the president. The administration's hope is that an improved foreign public opinion climate can be used as a resource in negotiating with the leaders of the other nation, to increase the likelihood that the president will realize his foreign policy goals vis-a-vis the visited nation. An underlying assumption of high-level public diplomacy appears to be that foreign leaders are responsive to public opinion pressures within their own country, like members of Congress and the president are thought to be responsive to U.S. voters.

Currently, there is only a limited literature on high-level public diplomacy. Several early studies have looked at what was then termed presidential "going international" (Rose 1988; Smith 1997). These studies focused on the logic of why presidents increasingly go international in the post-Cold War era (Rose 1988) and tracked trends in presidential going international activities like foreign trips (Smith 1997). A more recent literature has turned its attention to the effectiveness of high-level public diplomacy on foreign public opinion and support for U.S. policies. Several studies find that such public diplomatic efforts can affect foreign public opinion, at least under some conditions (Dragojlovic 2011, 2013; Goldsmith and Horiuchi 2009). Furthermore, the climate of public opinion in the foreign nation is associated with greater support for U.S. positions on foreign policy issues, for instance, on roll-call voting in the United Nations (Datta, 2009; Goldsmith and Horiuchi 2012).

One of the major barriers to assessing the effectiveness of high-level public diplomacy, especially on public opinion, is the dearth of cross-national data on public opinion about the United States, its policies, and the president.³ Moreover, the extant literature has not investigated all of the causal steps from a presidential visit to foreign public opinion. This article asks whether a presidential visit to a nation can focus public attention on the president.⁴ The agenda-setting (McCombs and Shaw 1972) and political communication (Edwards 2006; Zaller 1992) literatures argue that prior to the president being able to influence public opinion targeted citizens must receive the president's communication. Can the president focus foreign public opinion on his communication to them? To test the presidential international attention focusing hypothesis, I construct a pooled cross-section time series of weekly Google searches for Barack Obama across forty-two nations during the president's first term. Analysis finds that, during the week of the president's visit, such searches increased by about 25%, indicating relatively potent attention focusing effects from a presidential visit.

This article proceeds as follows. First, I discuss agenda-setting and political communication theories as they pertain to presidential attention-focusing activities. Then I review the literature on high-level public diplomacy by American presidents and discuss why presidential trips to another nation should focus local public attention on the president. The following section presents the Google Trends data, which is followed by the analysis. The conclusion puts the findings into perspective and raises suggestions for future research.

3. See the efforts that Goldsmith and Horiuchi (2009) go through to create such an indicator.

4. The term "attention focusing" was first coined by Lammers (1982).

Agenda Setting, Presidential Communications, and Attention Focusing

Early research on agenda setting has looked at the impact of news coverage on the public's issue priorities. That research found that when news organizations increase their reporting on specific issues or topics, those issues and topics would rise in the public's priority rankings (McCombs 2013; McCombs and Shaw 1972, 1993). Later research asked whether other communications, such as major presidential speeches like the State of the Union Address, can also affect the public's issue priorities (Cohen 1995; Eshbaugh-Soha and Peake 2011; Hill 1998). That research found that presidential public communications can influence public issue priorities but not issue preferences (Edwards 2006).

Priming appears to be one mechanism that accounts for these communication effects on the public's agenda. For a presidential message to affect public opinion, the voters must receive the message (Edwards 2006; Zaller 1992). Voters may learn about the president's speech directly, for instance, by watching it. They may also learn about the president's speech indirectly, from conversations with family, friends, coworkers, and so on. (Cohen 2010; Edwards 2006; Zaller 1992). Even with these two transmission mechanisms from the presidential speech to the vote, as Edwards details, in modern American politics, there are many barriers to voters receiving presidential communications (Edwards 2006; see also Baum and Kernell 1999; Kernell and Rice 2011). For instance, Young and Perkins (2005) show that as the audience for major presidential addresses has decreased in size, the effect of the president's public rhetoric on the public's issue agenda has also weakened. Contemporary presidents seem less able to affect the public agenda of American voters than was the case a generation ago.

In contrast to public communication from the president to domestic audiences, which are frequent and routine, a presidential trip to a foreign nation is rare and special. Presidents do not travel to each nation every year, and most nations will receive at most one visit during the entire time a president is in office. Because such trips by the president to the overwhelming number of nations are rare, the news media in the visited country will report on the trip with high volumes of news coverage. Politicians and other elites in the visited nation may also want to take credit for the president visiting their nation and perhaps have some of the prestige of an American president passed onto them. To do so, local political leaders, especially heads of state, may orchestrate events that show them in the president's company. These attributes of a presidential trip to a foreign nation increase the likelihood that large numbers of citizens in the visited nation will become aware of, and perhaps even interested in, the president's visit. This broadscale awareness lays the foundation for potential presidential influence over foreign public opinion.

High-Level Public Diplomacy and Attention Focusing of Foreign Public Opinion

There is empirical evidence that, under certain conditions, high-level public diplomacy by the president, and the secretary of state, can influence foreign public opinion.

Goldsmith and Horiuchi (2009), for example, find that presidential, and secretary of state, visits are associated with a more positive image of the United States, as long as the public in the host country views the United States as credible and trustworthy.⁵ Once that credibility begins to flag, visits may no longer influence foreign public opinion. And if the United States or the president is viewed as “noncredible,” high-level diplomatic visits may actually heighten negative attitudes toward the United States (Goldsmith and Horiuchi 2009). Dragojlovic (2011) employs an experiment on Canadian college students, adding support to the Goldsmith and Horiuchi (2009) perspective. When presented with an Obama prime shortly after his inauguration, when Obama enjoyed high credibility, treated subjects in the experiment displayed more positive attitudes toward the United States than nontreated subjects. However, if exposed to a George W. Bush prime, when Bush’s credibility was weak, attitudes toward the United States became more negative.

There is also some empirical support for the second claim of the high-level public diplomacy literature, that foreign public opinion will affect the nation’s foreign policy. Datta (2009) and Goldsmith and Horiuchi (2012) find that support for the U.S. position on issues of interest to the United States, like support for the Iraq War and voting patterns in the United Nations, is positively associated with public opinion about the United States.

But as noted above, prior to a high-level public diplomatic activity affecting public opinion, citizens in the visited country must receive the president’s communication. Can a presidential trip to another nation effectively compete with all of the other stimuli that vie for the attention of citizens in that host nation? Where the president’s ability to focus public attention appears to be waning within the United States, we expect that trips to other nations are able to focus the attention of foreign publics on the president and his policy objectives in visiting the nation.

First, although presidents travel to foreign nations frequently, they rarely visit any nation more than once a year. Still, presidents do visit a handful of nations multiple times per year. Generally, these nations are powerful in international politics and are often important U.S. partners on critical issues such as the United Kingdom, Germany, and/or France. The international power and importance of these nations leads presidents to visit them with some frequency. Similarly, the leaders of these nations are likely to visit the United States several times per year for much the same reasons that the president visits them, indicating a high degree of high-level interactions between the United States and these nations. Presidents may also visit less powerful nations more than once a year, primarily when these nations are involved in timely and important issues that affect the national security and/or other interests of the United States. South Korea and Indonesia are two such examples. South Korea is strategically important to the United States, while

5. Goldsmith and Horiuchi (2009, 865) distinguish presidential credibility from the president being controversial or noncredible. A credible president is one who is trustworthy, a noncredible one is a president who is not trustworthy, while a controversial president is one who takes positions that rouse debate and opposition around the world. The credible–controversial–not credible distinctions come from the literature on messengers attributes in communications.

Indonesia is becoming an economic power in a region that has been gaining increased attention from the United States.

Table 1 presents a breakdown of the number of presidential visits to forty-two nations during President Obama's first term of office (2009-2012), nations for which we have weekly Google search data. France was the most frequently visited nation, with five visits during those four years, while the president visited the United Kingdom and Japan three times each and Germany twice. President Obama visited South Korea three times, neighboring Mexico three times, and Indonesia three times, and several other nations received two visits from the president during his first term.

With only from one to two dozen presidential international trips per year, most nations do not receive even a single presidential visit in any given year. A number of reasons may lead a president to visit one of these nations, at least occasionally. For instance, a specific issue may have arisen that requires attention, a new regime has come to power, the president wants to establish a better working relationship with the leader of the host nation, and/or the president has not visited the nation for an extended period of time. There are many other, and perhaps idiosyncratic, reasons that motivate a president to visit any particular nation from this set of infrequently visited nations. Of the forty-two nations on Table 1, twenty did not receive a presidential visit during Obama's first term in office, and another ten were visited only once. Presidential visits to foreign nations, although growing in the aggregate over time, are still rare events for most countries most of the time.

The rarity of a presidential visit may lead politicians and citizens of the visited nation to view the president's visit as important and noteworthy. If citizens view a president's visit to their nation as special and important, it is likely that the president will be able to focus public attention on himself and the international issues of interest to the president. In this sense, a presidential visit to a foreign nation may be comparable to a prime-time presidential television address to U.S. voters. Presidents rarely broadcast to the United States on television during prime time, doing so only when the issue is both timely and important (Foote 1990; Kernell 1993). A presidential prime-time address to the nation signals to voters that the president thinks something is important and wants the public to pay attention to the president and the issue. Although presidential trips to another nation do not necessarily involve critical issues, as is often the case for a domestic prime-time address, the infrequency of international visits may make them important in the eyes of foreign citizens.

Hence, we can think of a presidential trip to a foreign nation, and the public activities that he performs while on the trip, as a presidential attention-focusing activity. The question here is: does the president's trip focus the attention of the foreign public on the president? To test this idea, we need data on foreign public opinion and/or behavior. The next section discusses the data used in this article, derived from Google search behavior.

Measuring Public Attention with Google Trends Data

One of the aims of a presidential public activity is to alter public opinion. In the international context, presidents may want to affect foreign public opinion about the

TABLE 1
Descriptive Statistics on Google Searches for “Barack Obama” and the Number of Presidential and Secretary of State Visits, by Country, February 2009 through October 2012

Country	Barack Obama Searches				Visits	
	Mean	Std. Dev.	Min	Max	President	Secretary of State
Argentina	15.79082	13.37898	0	100	0	1
Australia	23.79082	11.43912	9	100	1	2
Austria	26.75	13.98291	9	100	0	0
Belgium	27.64286	13.35953	9	100	0	6
Brazil	12.11224	10.89567	4	100	2	4
Canada	23.72449	13.57999	10	100	2	4
Chile	7.311224	7.664776	0	100	1	1
China	13.9898	12.05117	0	100	1	7
Colombia	13.7398	11.87795	0	100	2	1
Czech	4.632653	8.23985	0	100	3	0
Denmark	16.39286	11.1587	0	100	0	2
Finland	32.5102	17.15584	0	100	0	1
France	32.19898	15.54567	12	100	5	5
Germany	28.36224	13.91463	10	100	2	7
Hungary	22.67857	13.68262	0	100	0	1
India	8.311224	8.012175	3	100	2	3
Indonesia	10.12755	13.92688	2	100	3	4
Ireland	3.443878	7.284822	1	100	1	1
Israel	18.22959	18.6488	0	100	0	5
Italy	9.255102	8.504371	3	100	1	1
Japan	20.14796	17.01714	0	100	3	5
Malaysia	30.07653	17.97518	0	100	0	1
Mexico	13.94388	9.875217	5	100	3	4
Morocco	13.13265	16.03595	0	100	0	2
Netherlands	25.11224	12.87485	7	100	0	2
New Zealand	38.55102	18.81959	0	100	0	1
Nigeria	20.63776	18.9476	0	100	0	2
Norway	8.831633	8.656832	2	100	1	2
Peru	24.45408	16.26804	0	100	0	2
Philippines	10.46939	10.10479	3	100	0	2
Poland	5.806122	7.954561	2	100	1	1
Portugal	12.87755	9.042651	4	100	1	0
Romania	13.86224	15.2257	0	100	0	0
Saudi Arabia	21.26531	27.07314	0	100	1	2
South Africa	21.84694	13.33348	6	100	0	3
South Korea	14.20918	22.58391	0	100	3	6
Spain	27.15306	14.02734	4	100	0	1
Sweden	32.64286	17.06293	9	100	0	1
Switzerland	27.30102	14.21001	11	100	0	5
Turkey	4.928571	7.354399	0	100	1	8
United Kingdom	20.53061	10.19674	8	100	2	5
Venezuela	20.43367	20.34751	0	100	0	0

Source: Google Trends for Barack Obama searches and the Office of the Historian of the State Department for President and Secretary of State Visits, <http://history.state.gov/>.

United States, the president, and/or a particular U.S. foreign policy issue. By producing a more favorable foreign opinion climate, the president's negotiating position with the leader of that nation may be improved, enabling the president to obtain a policy response from the foreign leader closer to the president's foreign policy goal than would result with a less favorable opinion climate.

The attention-focusing hypothesis tests whether the target (foreign public opinion) receives the message (the president's public activity) from the messenger (the president). At the individual level, for the president's message to affect opinion, the target must receive or become aware of the message. The targeted individual(s) may receive the message either directly or indirectly. An individual may receive the president's message directly, for instance, by watching a television broadcast of a presidential speech and/or by reading about the speech in a newspaper. Individuals also may receive the message indirectly, for instance, by being informed about the president's speech from a family member, coworker, or friend. Through this indirect route, a president's message potentially can affect opinion of a large number of people even if not many people learned of the speech directly. Message reception, however, does not guarantee that the president's speech will influence opinion; message reception is merely a *necessary* precondition to opinion change. If an individual does not receive a message, either directly or indirectly, the message cannot affect his or her opinion.

Generally, public opinion polls are used to measure message reception and opinion change, but there are several complications in using public opinion polls to test for presidential leadership effects on opinion when visiting other nations. First, there is often a temporal mismatch between visits and polls—for some nations there may be visits but no polls and for others, polls but no visits. Rarely do we have polls being administered in a foreign nation that coincide with the president's visit. Second, as Goldsmith and Horiuchi (2009) point out, polling attitudes about the United States, and polling in general, is not randomly distributed around the world. It is more common in more well-developed nations such as in Western Europe. Thus, there is a selection bias to the nations for which we have polling data. This type of selection bias may affect statistical estimates.

Third, when there are relevant polls in the visited nation, there may be a large time gap between the visit and the poll. The larger the time between the visit and the poll, the more likely that a factor besides the presidential visit may be affecting public opinion such as other local or international events. Moreover, the longer the time gap between the visit and the poll, the more likely that the effects of a visit on opinion may have dissipated. In their study, Goldsmith and Horiuchi (2009) find the effects of a high-level visit on public opinion appear to persist for up to five or six weeks. This may appear to be quite a long time for visit effects to persist, but the effects of a presidential visit may persist if the news media and local leaders publicly discuss the issue raised by the visit in the following days and weeks and if the presidential visit set the public and political agenda.

Finally, that presidential visits are rare makes it difficult to locate matching polls. Trips by the secretary of state around the globe are more common. For instance, in the 2000s, the secretary of state consistently traveled to fifty or more nations per year. In Goldsmith and Horiuchi (2009), of 353 country polls that they gathered, there are only thirty-eight high-level visits within the five-week lag that they use in estimating their

statistical model. And of these thirty-eight visits from 2001 (after 9/11) through 2006, there are only fourteen for the president.⁶ To conduct their analysis, Goldsmith and Horiuchi combine presidential and secretary of state visits into one variable. By combining presidential and secretary visits, which is reasonable given the relative rarity of presidential visits, we cannot test whether a presidential visit has stronger effects on foreign public opinion than a visit by the secretary of state. Despite these limitations, matching visits and poll data is important and useful, but there is still much that we do not understand about the effects of a high-level visit on foreign opinion. In addition to the limitations noted above, the polls used in research on foreign attitudes toward the United States, such as in Goldsmith and Horiuchi (2009), lack measures on respondent awareness and/or exposure to high-level U.S. visits.

For these reasons, to test the presidential foreign public's attention focusing hypothesis, I use Google Trends data. Research has increasingly employed Google Trends data, especially for making forecasts in fields as diverse as disease outbreaks, economic behavior, and politics.⁷ Google Trends data have not always proved accurate for forecasting. But rather than forecasting, this article uses Google Trends data as a behavioral indicator for hypothesis testing: do we find an increase in Google searches for the president in response to a foreign visit?

Google Trends is a database of searches on Google, the most frequently used Internet search engine. One can retrieve the Google Trends data for specific geographic regions, such as a country, as well as time periods. After experimenting with several search terms, I selected "Barack Obama" for the period from February 2009 through October 2012. I limited the search to this time period because search volume for "Barack Obama" spiked during November 2008 to January 2009, and again in November 2012, a function of the presidential election contest and inauguration in January 2009. Google Trends data are presented as the percentage of searches compared to the peak search. The peak search is given the value of 100 in the Google Trends report. For privacy, competitive, and other reasons, Google does not release the actual number of searches. The volume of searches during the election/inauguration periods was extremely high, even outside of the United States, compared to other periods, making it difficult to discern variance during these other periods because of Google's reporting practices—Google only reports whole numbers, such as a "0," "1," "10." By restricting the search period as described, we are better able to see the peaks and valleys in searches for "Barack Obama."

To see if a presidential trip affected the relative number of searches, I searched "Barack Obama" for each nation separately, only using data that were disaggregated by week. Google Trends provides weekly data when the volume of searches is large enough to allow such a refined disaggregation; otherwise it presents the data in monthly units.⁸ The weekly data are temporally refined enough for testing the attention-focusing

6. The secretary of state always accompanies the president when the chief executive visits another nation but may visit a nation alone, without the president.

7. For studies that use Google Trends data, see Choi and Varian (2012), Pelat et al. (2009), and Vosen and Schmidt (2011). For applications to political data, see Lui, Metaxas, and Mustafaraj (2010), Reilly, Richey, and Taylor (2012), and Weeks and Southwell (2010).

8. If enough search volume exists, Google Trends presents the data in daily units.

hypothesis. This short, weekly time unit will increase confidence that the visit, and not some other event, stimulated the search volume. Wider time units, like months, may be too broad for us to isolate presidential trip effects. Other events/processes may occur within a month that might affect foreign public attention to the U.S. president. Finally, the way Google makes the data public requires using the same search terms and time spans to retrieve comparable data. Excluding the United States, I was able to collect this weekly data for forty-two nations for 196 weeks, for an n of 8,232 in country-week units. The nations, along with descriptive information on country-specific search patterns, are listed on Table 1.

Unlike the opinion data gleaned from polls, the Google Trends data provide us with a behavioral measure, search behavior. Still, there are limits to these data. The Google Trends data are not based on representative samples, nor are the search data completely comparable across countries. First, to search on Google requires Internet access, which is not randomly or equitably distributed within nations, and Internet penetration varies across nations, with higher penetration levels in more advanced nations. But Internet searching is not an especially difficult or costly activity and has become increasingly commonplace for large segments of the population. Second, of those individuals with Internet access, only a nonrandom subset will use a search engine, and an even smaller, and perhaps more unrepresentative, group will search for "Barack Obama." Notably, the search traffic volume in these forty-two nations was high enough for Google to quantify in weekly units. Third, without demographic and other data on who searches and for what reasons, we have to be careful in making comparison statements across nations. Lastly, the Google Trends data only tell us something about interest in "Barack Obama" by searchers, not by citizens at large. These data do not convey information on attitudes toward Obama such as whether they like or dislike him, agree or disagree with his policies, and so on. Still, given these caveats, we can test the attention focusing hypothesis across a large number of countries over a relatively sustained period of time with comparable data, which enables us to address the following question: are there more searches for Barack Obama during the week when he visited a country than for weeks without visits?

Estimation Issues

With data across nations and time, the analysis employs pooled cross-sectional time series techniques. With a relatively large number of panels (forty-two nations) and 196 time units (weeks), both temporal and cross-sectional disturbances may affect results (Beck and Katz 1999, 2011; Eberhardt 2012; Franzese and Hays 2007; Wilson and Butler 2007). It is highly likely that these data will exhibit cross-sectional correlations, that is, search interest in "Barack Obama" in one nation will affect similar searches in other nations. First, there is now a global news media, such that major events transpiring in one locale are reported around the globe. Many presidential trips are of interest not only in the nation visited, but in other nations, and presidential international travel tends to be reported by the global media (Farnsworth, Lichter, and Schatz 2013). Thus, a

presidential trip in one country may spark interest among citizens in another country, leading to increased search volume beyond the visited country. This spillover means that cross-sectional observations in these data are not necessarily independent of each other, a requirement for statistical analysis.

I employ several diagnostics to test for these issues in these Google search data. First, the Pesaran CD statistic tests for cross-section dependence in panel time-series data (Pesaran 2004). Essentially, the Pesaran test estimates the cross-sectional correlations of each panel with all the others and then presents a summary statistic (CD), which can be thought of as akin to an average correlation. The CD test statistic of 153.44 ($p < 0.000$), with an (average) correlation of 0.37 strongly suggests cross-sectional correlation in these data. I also use the Breitung test for unit roots in panel data (Breitung and Pesaran 2008). The Breitung robust lambda statistic of -2.89 ($p = 0.002$) does not indicate unit roots in the series. Further, the robust lambda statistic, with four lags, a trend term, and the series demean, of -4.14 ($p = 0.000$), reinforces that interpretation.⁹

Analysis

To test the presidential attention-focusing hypothesis, I regress the weekly searches for “Barack Obama” on a presidential visit variable, scored one if the president visited the nation during that week and zero otherwise. The analysis also controls for several other variables. The first is a dummy variable for whether Secretary of State Hillary Clinton visited the nation in a solo trip, that is, without the president (1 = secretary of state visit, 0 = no visit). It is possible that a visit from the secretary of state will affect search interest in the president. Also, I added this variable to the estimated model because Goldsmith and Horiuchi (2009) combined presidential and secretary of state trips into one variable in their measure of high-level public diplomacy. The analysis also controls for the announcement of the killing of Osama Bin Laden, an event that received worldwide attention as well as credit for President Obama. This variable is scaled one for the week of that announcement, zero otherwise.¹⁰ A simple comparison indicates an increase in searches for Barack Obama during this announcement week (44.3) compared to other weeks (18.4). Finally, the estimation controls for general searches for the “United States.” This variable is intended to pick up other factors that may stimulate someone to search for Barack Obama, perhaps as a byproduct or spillover of searching for the United States.

Table 2 presents an initial estimation, which clusters on country, the clustering meant to control for country-specific effects. All variables are significant predictors of search levels for Barack Obama except for solo secretary of state visits. The coefficients indicate that during the week of the announcement of Bin Laden’s killing, searches for “Barack Obama” rose by 18.4%, in line with the simple comparisons presented above. Second, general searches for the “United States” also led to increased searches for the

9. The robust lambda accounts for the cross-sectional correlation. Demeaning subtracts the cross-sectional means from each series.

10. The week of May 1-7, 2011.

TABLE 2

Impact of Presidential Trips on Weekly Google Searches for "Barack Obama," February 2009-October 2012

VARIABLES	Model (1)	Model (2)
	Regression Clustered on Country	Panel Corrected Standard Errors
Presidential Trip	42.99*** (5.493)	37.21*** (1.228)
Secretary of State Trip	1.160 (1.125)	0.895 (0.761)
Bin Laden Announcement	18.37*** (3.419)	17.90*** (2.235)
Searches for "United States"	0.436*** (0.0359)	0.342*** (0.0231)
Constant	6.696*** (0.765)	9.552*** (0.917)
Observations	8,232	8,232
R ²		0.273
R ² Overall	0.202	
R ² Within	0.285	
R ² Between	0.051	
sigma_u	8.737	
sigma_e	12.04	
rho	0.345	
Wald chi2/p	375.8 0.000	1256 0.000

N = 8,232 Groups = 42, Time Units (weeks) = 196. Robust standard errors in parentheses.

*** $p < .01$, ** $p < .05$, * $p < .1$.

Source: Google Trends for Barack Obama searches and the Office of the Historian of the State Department for President and Secretary of State Visits, <http://history.state.gov/>.

president. The coefficient suggests that each 1% increase in searches for the "United States" leads to 0.44% more searches for "Barack Obama." Searches for the "United States" in these data range from 12 to 100, with a mean of 26.4 and a standard deviation of 15.5. A one-standard-deviation shift in general "United States" searches will produce a shift of about 6.8% in searches for "Barack Obama."

Most important for this research is the positive effect of a presidential trip on searches for "Barack Obama." The coefficient indicates that during the week of a presidential trip, searches increased by a massive 43%. We should be suspicious of this massive impact, however. As discussed above, factors such as temporal dependence in the dependent (and other variables), as well as cross-panel correlation, may affect estimation. Does the finding that a presidential trip stimulates increased searches for the president hold when using other estimation techniques that can account for these statistical issues?

As a first test, Table 2 also presents results of an estimation using panel corrected standard errors. Panel corrected standard errors assume that the disturbances in the data are heteroskedastic and correlated across panels. The specific correction that I used assumes AR1 processes for each panel, but that the coefficient for each AR1 process is specific to each panel. Inspecting the results for this estimation on Table 2, the first thing to

notice is that the rho statistic falls short of statistical significance, indicating that correcting for autoregression is not necessary.

Yet, the results closely resemble those for the cluster effects estimations (Model 1) on Table 2. Again, all variables are significant, except for the secretary of state trip dummy, as before. Each of the significant independent variables exhibits slighted smaller regression coefficients but much larger significance and z-score values. The inflation in significance tests to such high levels often results from applying the panel corrections and thus may not be the proper specification for these data. I have presented the panel corrected standard error estimation because that approach is so common in political science research (Beck and Katz 1995, 2011; Wilson and Butler 2007). Other estimations techniques may be more appropriate for the data used here, especially for dealing with the cross-sectional correlation, which appears to be a consequential concern based on the above diagnostics.

Table 3 presents results of two estimations that better account for cross-panel, or spatial, correlation, the Pesaran Common Correlated Effects Mean Group (CCEMG) estimator, and the Eberhardt and Teal Augmented Mean Group (AMG) estimator (Pesaran 2006). The CCEMG estimation deals with cross-sectional correlation by including cross-sectional averages of the dependent and independent variables as regressors in the estimation. Although the regression coefficients for these averaged variables are reported, they do not have a meaningful interpretation. Where the CCEMG considers the averaged variables as only having relevance for estimating the equation, the AMG estimator represents a cross-dynamic process that is shared across the panels, in this case, country.¹¹ For purposes here, we are not concerned with these estimators other than their ability to correct for cross-panel correlation, and that cross-panel correlation may be affecting the effects of the substantive variables in the analysis.

Results on Table 3 generally parallel those in Table 2 in substantive terms, except that the regression coefficients and z-scores are smaller, albeit still significant. Again, results indicate that visits by the secretary of state do not affect searches for “Barack Obama.” Moreover, the announcement of the Bin Laden killing is no longer significant. As interest in President Obama peaked worldwide due to the assassination, it makes sense that the CCEMG and AMG estimators, which are meant to pick up properties common across panels, would do so for the Bin Laden variable, because searches for the president spiked worldwide as a result of the announcement. The two estimations, however, differ in the effects of general “United States” searches, with the CCEMG relegating that control variable to statistical insignificance, while the AMG estimation finds it to be significant, healthily so with a z-score of over 12. This variable however is of limited theoretical interest here and was used mainly as a control.

Both estimations however find that a presidential trip boosts searches for the president at statistically significant levels, repeating the above results. The z-score is the same for both estimations (4.75), although it is considerably lower than found above, but still clearly significant. Plus, both estimations report similarly sized regression coefficients,

11. See Pesaran and Smith (1995), Pesaran (2006), and Eberhardt and Teal (2010) for technical discussions of these estimators.

TABLE 3

Impact of Presidential Trips on Weekly Google Searches for "Barack Obama," February 2009–October 2012, Corrected for Cross-Panel Correlation

<i>VARIABLES</i>	<i>Model (3)</i> <i>CCEMG Estimator</i>	<i>Model (4)</i> <i>AUG Estimator</i>
Presidential Visit	24.64*** (5.186)	24.71*** (5.205)
Secretary of State Visit	0.319 (1.198)	0.699 (1.178)
Bin Laden Announcement	0.274 (3.944)	14.45*** (3.452)
U.S. Search	0.00228 (0.0351)	0.660*** (0.0528)
"Barack Obama" Searches Average	0.995*** (0.0989)	
Presidential Visit Average	−39.03*** (12.13)	
Secretary of State Visit Average	−0.605 (5.287)	
Bin Laden Average	0 (0)	
U.S. Searches Average	0 (0)	
cdp+		0.991*** (0.0964)
Constant	0.0257 (1.023)	0.683 (0.949)
Wald chi2	22.88	462.6

$N = 8,232$, Groups = 42, Time Units = 196.

Standard errors in parentheses *** $p < .01$, ** $p < .05$, * $p < .1$.

+ cdp refers to the "common dynamic process"

Source: Google Trends for Barack Obama searches and the Office of the Historian of the State Department for President and Secretary of State Visits, <http://history.state.gov/>.

about 24.6 to 24.7. These effects are about 12% less than found in the panel corrected standard error estimation on Table 2 (Model 2) but still substantively impressive. Based on these results, when a president visits a foreign nation, the level of Google searches for the president increases by approximately 25% compared to a week when the president did not visit the nation.

Finally, I estimate an Error Correction Model (ECM) for panel data that employs the mean group averaging to deal with cross-panel correlation, an estimation that allows for correction for temporal as well as cross-sectional issues. In this estimation, the variables are expressed as changes, with an ECM mechanism that picks up long-term trends. Table 4 presents these results. The ECM mechanism is positive and significant ($b = 0.39$, z -score = 11.34). Again, foreign visits by the secretary of state have no impact on searches for the president, but the Bin Laden dummy and general "United States" searches do. The results report that searches for "Barack Obama" increased by nearly 16% during the week

TABLE 4
Impact of Presidential Visits on Weekly Google Searches for “Barack Obama,” February 2009-October 2012

VARIABLES	(5) Error Correction, Mean Group Averaging Estimation
Error Correction Mechanism	0.389*** (0.0343)
Presidential Trip (changes)	23.51*** (4.862)
Secretary of State Trip (changes)	0.587 (0.592)
Bin Laden Announcement (changes)	15.84*** (2.170)
United States Searches (changes)	0.330*** (0.0309)
Constant	−2.426*** (0.380)
Observations	8,190
Groups	42
Time Units	195
Sigma	62.17

Standard errors in parentheses *** $p < .01$, ** $p < .05$, * $p < .1$.
Source: Google Trends for Barack Obama searches and the Office of the Historian of the State Department for President and Secretary of State Visits, <http://history.state.gov/>.

announcing the assassination of Bin Laden and that an increase of 1% in searches for the “United States” corresponds with a 0.33% increase in searches for the president. Most importantly, presidential trips affect searches for the president. The coefficient indicates that changing from a week with no presidential trip to a week with a trip results in a 23.5% increase in searches for the president, an effect in line with that found for the cross-panel correlation analyses presented on Table 3. No matter the specification employed, all estimations indicate that when the president visits another nation, the percentage of Google searches rises. And depending on the specification employed, the effect ranges from 23 to 40%. These are substantively meaningful increases, as well as being statistically significant.

Conclusion

A growing literature has found that high-level public diplomacy by presidents, under some conditions, appears to affect foreign public opinion toward the United States and foreign policies toward the United States. The causal mechanisms from these public diplomatic efforts to public opinion have not been specified fully, although some linkages have been identified. First, it appears that the United States must be viewed as credible for a high-level diplomatic trip to enhance the image of the United States among foreign

publics (Goldsmith and Horiuchi 2009). This article investigates another linkage in the causal process from high-level diplomatic trips to foreign public opinion: message reception. As agenda-setting and communication theories argue, before a communication can affect behavior or opinion, the target must receive the message. Message reception is commonly tracked with public opinion surveys that ask respondents about awareness of events and actions of political leaders. Unfortunately, for the topic under investigation here, there are not enough surveys that coincide closely enough in time to a presidential trip for analysis. Without such data, it is hard to assess whether a person received the communication, but we can look at the behavioral and/or opinion implications of communication reception.

That is the tack taken in this article, which asked whether trips resulted in increased foreign public attention to the president, as tracked with Google searches. With data spanning forty-two nations across 196 weeks during Barack Obama's first term, the analysis found that searches for the president in the visited nation rise by about 25% during the week of the visit. These results support the notion that presidents can focus public attention through high-level public diplomacy activities and converge with other studies that find that such activities appear to affect actual opinions, at least under some conditions (Dragojlovic 2011, 2013; Goldsmith and Horiuchi 2009).

We did not find, however, that trips by the secretary of state also led to increased searching for the president. At first blush, this stands in contrast to Goldsmith and Horiuchi (2009). Due to the paucity of presidential trips, they combined presidential and secretary of state trips into one variable and thus did not distinguish the opinion effects of a presidential versus a secretary of state visit. Their modeling decision is reasonable given their data and the limited number of presidential trips to the nations during the time period that they studied. It is possible, though, for a visit by the secretary of state to change foreign opinion without altering search behavior for the president. A visit by the secretary of state may lead local political elites and the mass media to discuss the issue that promoted the visit. This heightened elite and media discourse may in turn affect citizen opinions.¹²

Still there are some notable limitations of the research reported here. First, it is for only one president across approximately a four-year period. Second, the Google search data do not tell us anything about why an individual searched for the president online or about whether the trip (and/or the search) affected the individual's attitudes toward the United States and the president. Third, Internet searches are not performed by a representative sample of citizens, so we cannot generalize the findings reported here to citizens in general. In fact, from the Google Trends data, we know virtually nothing about the individuals who searched for the president such as their demographics or motivations to conduct the search. Nor do we know if the search affected their level of information or their opinions. Still, Internet access and searching is becoming very widespread; hence, the results apply to a significant element of the mass public in these nations. Fourth, it is not clear exactly how or if Internet search behavior translates into opinion change, for the searchers as well as throughout the country. Information retrieved from a search may alter someone's opinion about the president, the United States, and/or the policies of the

12. Also, secretary of state visits may affect other kinds of search behavior such as for the secretary.

United States. Searchers may also communicate the results of their searches to nonsearchers, diffusing information gained from their search to others. The linkages among presidential trips, Internet searching, other expressions of attention focusing (like reading newspapers), and opinion change are as yet not understood fully. This article represents one attempt to fill in some of that gap and I hope will stimulate continued research on the effects of high-level public diplomacy.

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